

Overall Merit: Weak accept

Summary This is a 35-page vision paper from an NSF workshop, written by roughly 30 co-authors from academia and industry. The central argument is that AI and hardware must co-evolve through cross-layer co-design rather than being developed in silos. The authors set a target of $1000\times$ improvement in AI training and inference efficiency over the next decade, decomposed into $10\times$ from algorithms, $20\times$ from silicon, and $5\times$ from system-level gains.

The paper organizes the AI+HW stack into three layers (hardware technologies, algorithms/paradigms, applications/societal impact) and covers each in detail: bottlenecks like the memory wall and low cluster utilization (5–20%), opportunities like 3D integration and compute-in-memory, and action items for academia, industry, and government. A recurring theme is shifting the metric of success from FLOPs to “intelligence per joule.”

Strengths

1. Table 1 maps fine-grained technology layers (device/substrate, 3D integration, analog compute, photonics, memory, interconnect, compiler/runtime, algorithms) to their challenges and opportunities. This cross-cutting taxonomy is genuinely useful as a reference.
2. The paper is candid about obstacles. Section 5.4 names specific problems: misaligned academic incentives, the “everything is a GPU” brand inertia, legal barriers to distillation, and the U.S.–China power gap. Most vision papers stay vague about what could go wrong.

Weaknesses

1. The paper reads like a workshop report. Each section follows the same template (insights, challenges, Q&A, trends, pitfalls, action items), and the same points about co-design and memory walls appear in nearly every section. A tighter edit would have made the core claims land harder.
2. There is little quantitative backing for the $1000\times$ decomposition. Why $20\times$ from silicon and not $10\times$? For a paper arguing we should measure intelligence per joule, there are no joule-based measurements anywhere.

Detailed Comments The observation that real deployments run at 5–20% hardware utilization (Section 4.2) was the most compelling argument for why co-design matters more than peak specs. The Q&A sections are more interesting than the action item lists because they actually take positions, like the direct statement that attention is not all we need (Section 4.3, Q3).

Discussion Questions

1. The $1000\times$ claim assumes roughly multiplicative gains across layers. But these layers interact: algorithmic sparsity only helps if hardware supports sparse operations. How realistic is it to treat these factors as independent?
2. The paper frames intelligence per joule as the new metric. But this is task-dependent and hard to standardize. How would you actually build a benchmark suite around it?
3. The roadmap is U.S.-centric (NAIRR, NSF funding, U.S. power grid). How would the priorities differ for a country with abundant power but limited semiconductor access?